

Carry-No-Spill

Does Force Feedback Help Teleoperated Transport of Hazardous Liquids Under Visual Occlusion?

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Motivation

- Hazardous liquid transport is safety-critical in chemical, biomedical, and remote operation settings.
- In teleoperation, operators may not reliably observe internal fluid motion.
- Main purpose: **move fast while avoiding spill**

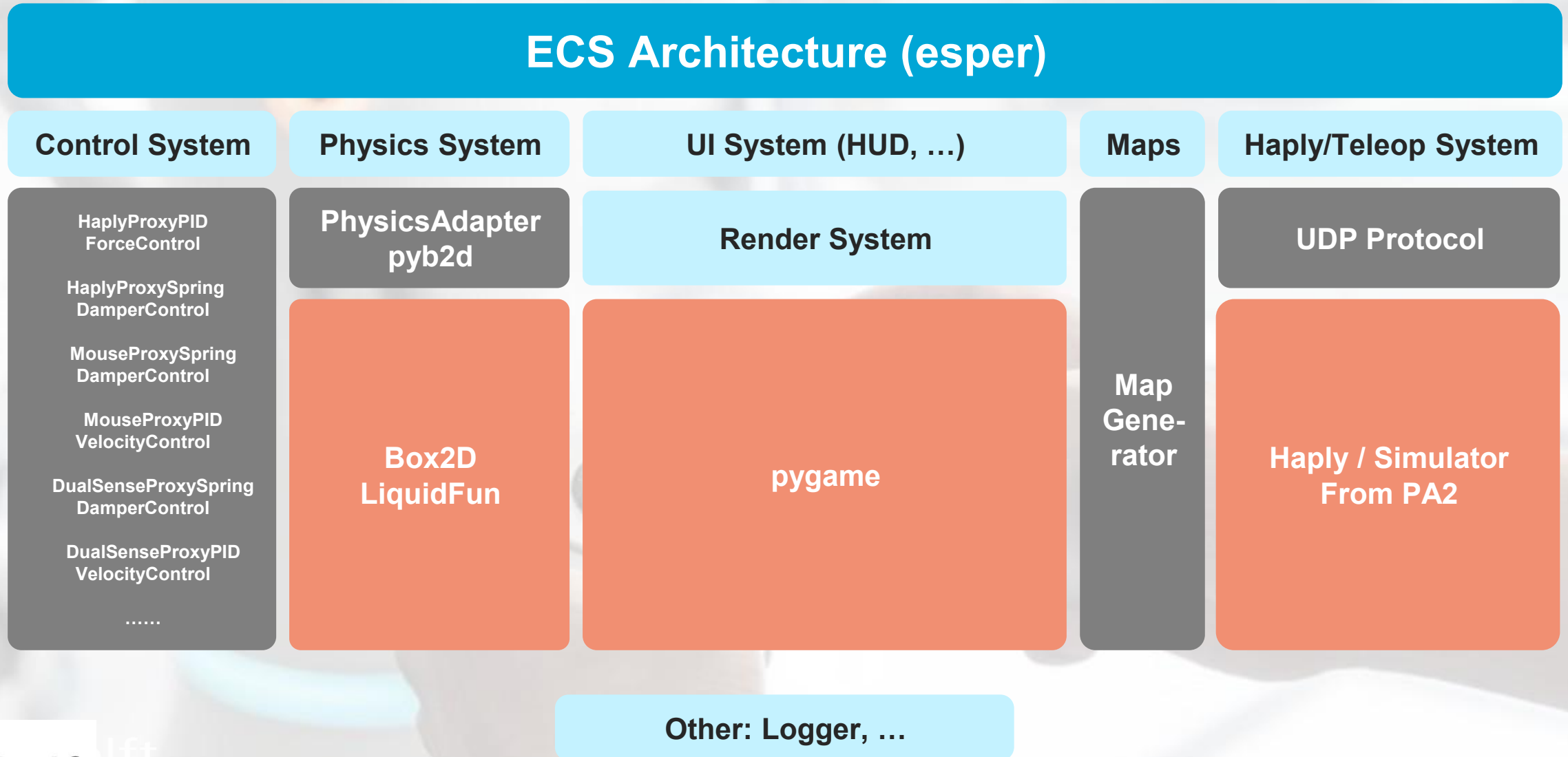


Research Question and Hypotheses

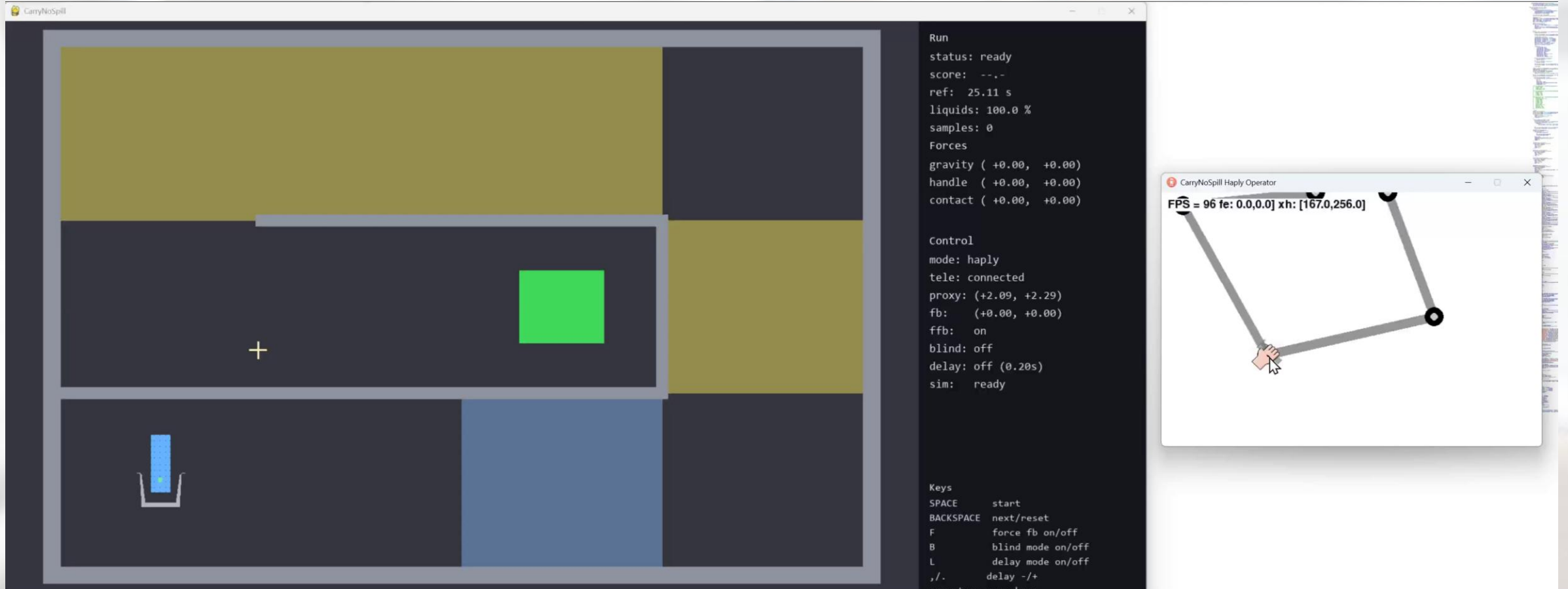
When fluid-state observability is reduced, does force feedback improve teleoperated carrying performance?

- $\mathcal{H}_0$: Force feedback does not improve performance over no force feedback.
- $\mathcal{H}_1$: Force feedback **improves** performance under blind transport conditions.

TODO Architecture and Simulation Platform



Demonstration



System and Controller Design

- Bidirectional teleoperation loop: operator motion in, feedback force out.
- Vessel is modeled as a dynamic rigid body containing liquid particles.
- Upright stabilization is achieved by control torque, not rigid angle locking.
- Force feedback is estimated from vessel dynamics and interaction forces.

$$m\mathbf{a} = \sum \mathbf{F}$$

$$F_{contact} = m\mathbf{a} - F_g - F_{handle}$$

$$F_{fb} = F_{contact} - F_{handle}$$

System and Controller Design

- A series of baseline controllers, e.g. **HaplyProxyPIDForceControl**

```
ecs.add_component(vessel, HaplyProxyPIDForceControl(  
    k_p = 7.6,  
    k_i = 1.8,  
    k_d = 0.06,  
    camera_entity = camera,  
    device_origin = self.settings.haply.device_origin,  
    k_angle_spring = 280.0,  
    k_angle_damper = 6.0,  
))
```

$$\mathbf{F}_{\text{cmd}} = m \left(k_p \mathbf{e} + k_i \int \mathbf{e} dt + k_d \dot{\mathbf{e}} \right) \mathbf{e} = \mathbf{x}_p - \mathbf{x}$$

$$\tau = I[k_\theta(\theta^* - \theta) - c_\theta \dot{\theta}]$$

HaplyProxyPID
ForceControl

HaplyProxySpring
DamperControl

MouseProxySpring
DamperControl

MouseProxyPID
VelocityControl

DualSenseProxySpring
DamperControl

DualSenseProxyPID
VelocityControl

.....

Experimental Design

Task: carry a liquid-filled beaker through a maze to the goal region.

In endless randomly generated mazes containing random walls, start / goal, and force fields.

Same seed and same runs for two different conditions.

Condition A: Blind & No Force Feedback

- water is invisible
- force feedback disabled

Condition B: Blind & Force Feedback

- water is invisible
- force feedback enabled

Metrics

Liquid score

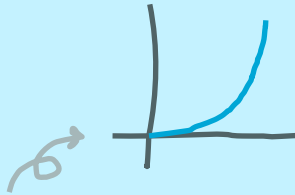
- Liquid retention ratio:

$$R_{liquid} = \frac{N_{inside}}{N_0}$$

- Liquid score:

$$S_{liquid} = S_{liquid}^{max} \times (R_{liquid})^{\gamma}, \quad \gamma > 1$$

(The upper liquid accounts for more points)



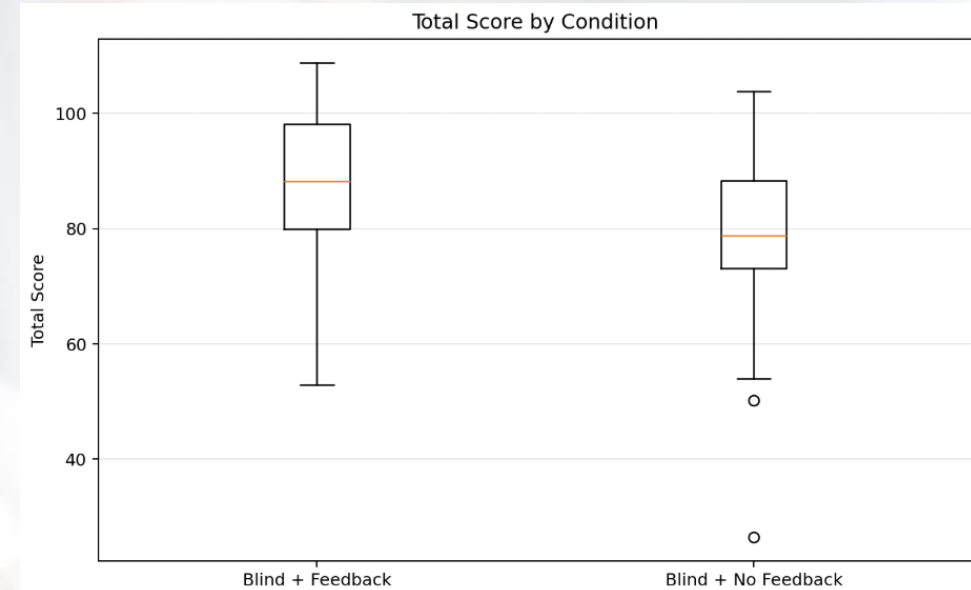
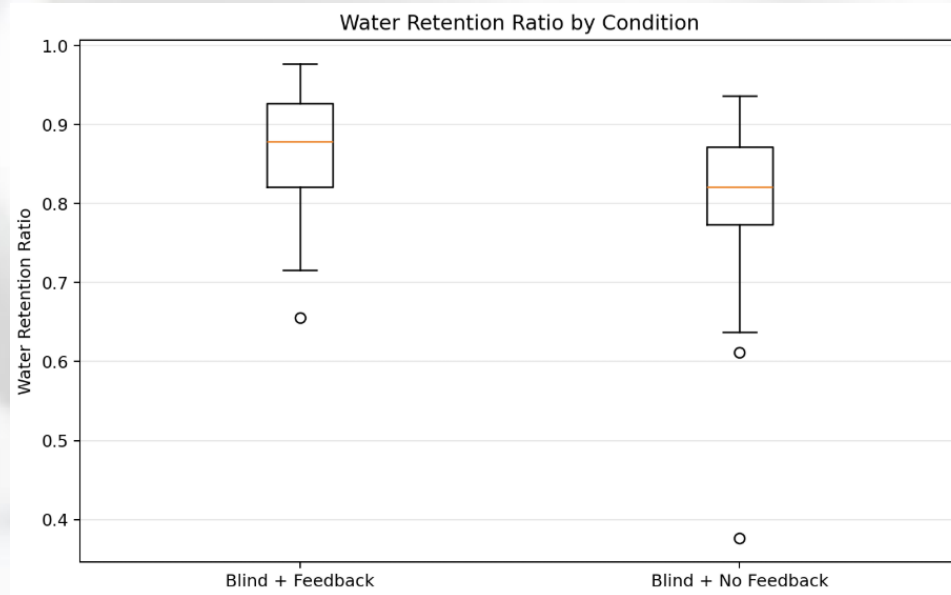
Time score

- Actual time used T ;
- Expected time consumption T_{ref} : calculated from the reference path, reference velocity, and number of force fields.
- Time score:

$$S_{time} = S_{time}^{max} \times \max\{0, 1 - T/T_{ref}\}$$

= Total score

Evaluation and Results



	with force feedback	w/o force feedback
Liquid score	<u>71.98 ± 14.97</u>	59.62 ± 16.81
Time score	16.13 ± 3.23	<u>17.89 ± 3.15</u>
Total score	<u>88.11 ± 13.56</u>	77.51 ± 16.53

* 29 runs per condition; 58 runs in total.

Statistical Interpretation

	Total score	Liquid score	Time score
Mean difference	+10.60	+12.36	−1.77
Welch t-test	$p = 0.0100$ $CI = [2.64, 18.56]$	$p = 0.0045$ $CI = [3.99, 20.74]$	$p = 0.0393$ $CI = [-3.44, -0.09]$
Cohen's d	$d = 0.70$	$d = 0.78$	$d = -0.55$
Mann-Whitney U test (robustness)	$U = 587.0$ $p = 0.0098$	$U = 599.5$ $p = 0.0055$	$U = 262.0$ $p = 0.0140$

We can reject the null hypothesis with $p = 0.010$, i.e.,
The improvement with the force feedback is **statistically significant**.

Discussion

Core Insights

Force feedback converts invisible fluid dynamics into actionable signals

- Makes hidden states perceivable
- Enables predictive (not reactive) control
- Improves safety under uncertainty

Conclusion

Under blind conditions, force feedback improves overall performance.

Future Work

Validation improvement:

- More environments (different seeds)
- Factorial study (Delay $\times$ FFB)
- Trajectory & spill analysis

Adaptive / informative force feedback

- Learn or design task-specific force encoding
- Amplify critical signals (e.g., instability, spill risk)
- Filter irrelevant dynamics